Regis University
College for Professional Studies
School of Computer & Information Sciences

SYLLABUS

COURSE NUMBER: CS 375

COURSE TITLE: Computation Theory

Course Description
Provides the concepts of languages and language analysis including lexical analysis, syntax analysis and regular expressions. Introduces automata theory including deterministic and nondeterministic finite state machines and parsing techniques. Introduces the concepts of NP and NP complete problems.

Course Prerequisites
MT320 – Introduction to Discrete Mathematics and

Course Outcomes
Upon successful completion of this course, students should be able to:

1. Explain the concept of language as used in theoretical computer science.
2. Explain and use regular expressions to represent languages.
3. Compare and contrast among regular expressions, finite state automata, nondeterministic finite automata, pushdown stack automata, and Turing machines.
4. Create and use finite-state, nondeterministic finite, pushdown automata & Turing machines.
5. Define and explain context-free grammars.
6. Define and explain the significance of P, NP, and NP-complete classes and the significance of tractability.
7. Describe the Chomsky hierarchy of regular, context-free, context-sensitive, and recursively enumerable.
8. Explain the Church-Turing hypothesis and Cook's Theorem.

Course Materials and Resources, including Technology Tools

Required Textbook:

Cohen, Daniel. (1997). *Introduction to Computer Theory, (2nd edition).* New York, NY: John Wiley & Sons, Inc. ISBN10: 0471137723 or ISBN13: 9780471137726.

Supplemental:

Method of Instruction

This course presents material using classroom lectures or online readings, textbook readings, and facilitated group discussions. Students are accessed via individual assignments and exams.

Related Standards

Related Standards	
Program Outcomes	
	This program enables students to achieve the following program outcomes by the time of graduation:
	An ability to apply knowledge of computing and mathematics appropriate to the discipline
	An ability to analyze a problem, and identify and define the computing requirements appropriate to its solution
	An ability to design, implement, and evaluate a computer-based system, process, component, or program to meet desired needs
	An ability to communicate effectively with a range of audiences
	An ability to analyze the local and global impact of computing on individuals, organizations, and society
	An ability to use current techniques, skills, and tools necessary for computing practice
	An ability to apply mathematical foundations, algorithmic principles, and computer science theory in the modeling and design of computer-based systems in a way that demonstrates comprehension of the tradeoffs involved in design choices
	An ability to apply design and development principles in the construction of software systems of varying complexity
Regis Nine Educational Outcomes	
	Content Knowledge in Discipline
	Diverse Cultures, Perspectives, and Belief Systems
	Arts, Sciences, and Humanities
	Critical Thinking
	Communication
	Technology
	Ethical and Social Responsibility
	Leadership in Service to Others
	Lifelong Learning
Jesuit Themes - These are themes that are embedded as part of our interactions and instruction at Regis, but are not written in as activities	
	Magis, Finding God in All Things, Unity of Hearts and Mind, Contemplatives in Action, Care of the Whole Person, Men and Women for Others

First Week Assignment

- Read Chapters 1 - 3 in the textbook.

Course Overview Grid

Week	Learning Topics	Activities	Reading Assignments	Assignments
1	Introduction to the Theory of Computation	Using languages	Ch. 1 – 3	Assignment #1 Due: Online: Midnight, Sunday Classroom: Before 2nd class
2	Regular Expressions and Finite Automata	Using regular expressions	Ch. 4 & 5	Assignment #2 Due: Online: Midnight, Sunday Classroom: Before 3rd class
3	Nondeterministic Finite Automata	Creating automata	Chapter 6 and pp. 135-140	Assignment #3 Due: Online: Midnight, Sunday Classroom: Before 4th class
4	Kleene's Theorem	Using * or + closure	Ch. 7, pp. 92-142	Assignment #4 Due: Online: Midnight, Sunday Classroom: Before 5th class
5	Regular Languages and Decidability	Using regular languages	Ch. 9, pp. 169-185 Ch. 11, pp. 207-217 Wikipedia (lookup): Cook-Levin theorem & NP-Complete	Assignment #5 Due: Online: Midnight, Sunday Classroom: Before 6th class **Midterm exam**
6	Nonregular Languages and Context Free Grammar	Using grammars	Ch. 10, pp. 187-196 Ch. 12, pp. 224-254 Ch. 13, pp. 259-265	Assignment #6 Due: Online: Midnight, Sunday Classroom: Before 7th class
7	Push Down Stack Automata	Stack demonstration	Ch. 15, pp. 289-312	Assignment #7 Due: Online: Midnight, Sunday Classroom: Before 8th class
8	Turing Machines	Trace Turing execution	Ch. 19, pp. 434, pp. 610 – 612	Assignment #8 Due: Midnight, Sunday **Final exam**

Student Evaluation Grid

Assignment	Value (percent of overall course grade)
Assigned homework/exercises	50%
Participation	10%
Midterm	20%
Final	20%
TOTAL	**100 %**

Regis University Grading Scale

A	93 – 100	C	73 - 76
A-	90 – 92	C-	70 – 72
B+	87 – 89	D+	67 – 69
B	83 – 86	D	63 – 66
B-	80 – 82	D-	60 – 62
C+	77 – 79	F	59 or Lower

Attendance/Participation
Facilitator will provide specific attendance/participation policies in accordance with the official attendance policy of the School of Computer and Information Sciences.

Attendance Policy of the School of Computer and Information Sciences
Student participation is essential. Students are responsible for all material covered in each week. Students' interaction is central to the learning experience and imparts critical observations and knowledge to all concerned.

If a student misses more than one week of class participation without prior faculty approval, the facilitator may assign a failing grade.

Late Assignments/Assignment Revisions
Facilitator will provide specific late assignments/assignment revisions policies not to exceed the maximum official policy of the School of Computer and Information Sciences.

Late Assignments/Assignment Revisions Policy of the School of Computer and Information Sciences
Occasionally, work, health, and family obligations have priority over course work. If you experience these issues, please notify the facilitator in advance. If an assignment is due, please make every effort to submit your assignment on time. In the case of a negotiated absence or delay, assignments turned in by the agreed upon due date will receive full credit.

If you have not negotiated with the facilitator, assignments turned in late will be graded, and then reduced by 3% per day. Assignments **will not** be accepted later than one week after due date. Week 8 assignments may only be 3 days late.

Incomplete Grade
An incomplete grade is given at the discretion of the facilitator. A facilitator is not required to assign an incomplete grade. Each occurrence will be negotiated on an individual basis.

Incomplete Grade Policy of the College of Professional Studies
A grade of Incomplete or 'I" denotes that the required work for the course is incomplete due to unforeseen circumstances. Unforeseen circumstances mean, for example, that an accident, an illness, a death, or a major life transition has occurred. This grade is awarded at the discretion of the instructor and is submitted with an alternate grade of "I/F."

A student must submit a written request to the instructor asking that an incomplete grade be assigned prior to the end of the term in which the course is taken. The course instructor will determine the length of time for course completion, and the remaining requirements to complete the course; however, the maximum time for completion is the end of the following semester.

Faculty will notify students in writing of the approval/denial of their request and provide instructions and guidelines required to resolve the incomplete grade. If the course work is not completed by the end of the next applicable semester and a grade change form submitted by the instructor, the incomplete grade reverts to the alternate grade assigned by the instructor and is calculated in the grade point average.

Tutoring Information:
Occasionally students need additional assistance with course content. Tutorial assistance is available to students through SmartThinking, writing assistance and personal tutoring. SmartThinking provides every student with ten hours of free online tutoring in writing, math, statistics, economics and accounting each year. Writing assistance is available in the form of Roving Writing Tutors and a variety of writing workshops. Individual personal tutors are available in a variety of discipline areas with fees and arrangements made between the individual student and tutor. For access to these services, go www.regis.edu , Current Student, Services for CPS Students, Academic Tools, Tutoring/SmartThinking.

Disability Statement:
Equal access to education means equal opportunity to learn in an academic setting. Under Section 504 of the 1973 Rehabilitation Act and the Americans with Disabilities Act of 1990, students with documented disabilities who are otherwise qualified have a right to equal access to classes at Regis University. If you have a documented disability requiring academic adjustments for this class, please contact the Director of the Office of Disability Services at **303-458-4941** for discussion of disability issues. Since accommodations are not retroactive, it is suggested that you contact the Disability Services Office as soon as possible; adequate lead-time may be required to receive accommodations, including review of appropriate documentation.

Academic Dishonesty
Regis University is committed to intellectual integrity in its academic pursuits. Sanctions may, therefore, be imposed by the College for Professional Studies faculty or schools for cheating (defined as using inappropriate sources of information on a test) or plagiarism (defined as presenting as one's own, the ideas, words, or products of another). Such sanctions may include a failing grade on the assignment, failure of the course, or expulsion of the student from the course or the department.

Non-Public, Confidential or Proprietary Information Compliance
For class projects that relate to a student's employment, the student is responsible for complying with all applicable policies of the employer, including those related to non-public, confidential or proprietary information